

CHAPTER 2

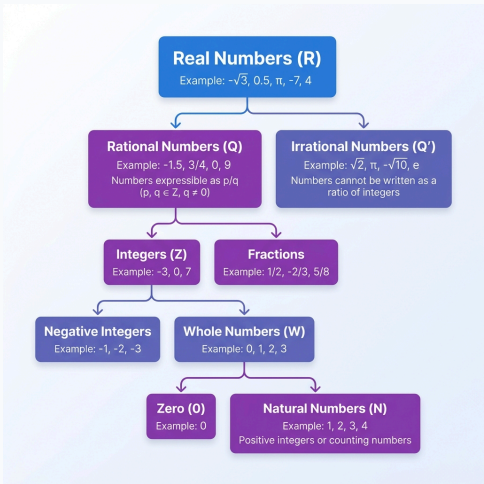
Real Numbers

Aligned with Samacheer Kalvi Class 9 Mathematics syllabus.

1. THE REAL NUMBER SYSTEM & RATIONAL NUMBERS

Numbers play a vital role in mathematics. In our lower classes, we started with counting numbers called natural numbers (N) and gradually expanded the system to include whole numbers (W), integers (Z), and rational numbers (Q).

Rational Numbers (Q): A rational number is a number that can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. The decimal representation of a rational number is either **terminating** or **non-terminating repeating (recurring)**.



NUMBER CATEGORY	DESCRIPTION & MATHEMATICAL SYMBOL	EXAMPLES
Natural Numbers	Counting numbers. Symbol: $\mathbf{N} = \{1, 2, 3, \dots\}$	1, 5, 23, 1000
Whole Numbers	Natural numbers along with Zero. Symbol: $\mathbf{W} = \{0, 1, 2, \dots\}$	0, 2, 8, 45
Integers	Whole numbers and negative natural numbers. Symbol: $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	-15, -3, 0, 9, 212
Rational Numbers	Numbers expressible as $\frac{p}{q}$ ($q \neq 0$). Symbol: $\mathbf{Q} = \{\frac{p}{q} : p, q \in \mathbf{Z}\}$	$\frac{3}{4}$, $-\frac{5}{2}$, 7 (which is $\frac{7}{1}$), 0.25 (terminating)

Period of Decimals

In non-terminating recurring decimal representation of a rational number, the repeating block of digits is called the **period** of the decimal, and the number of digits in the repeating block is called the **length of period**.

Example: For $1/7 = 0.142857142857... = 0.\overline{142857}$, the period is 142857 and the length of period is 6.

p/q Conversion of Recurring Decimals

Every recurring decimal can be converted to a fraction. The general method is to multiply by a power of 10 equal to the length of period and subtract to eliminate the decimal part.

Example: Express $x = 0.333... \Rightarrow 10x = 3.333... \Rightarrow 9x = 3 \Rightarrow x = 3/9 = 1/3$.

2. IRRATIONAL NUMBERS & NUMBER LINE

Irrational Numbers (Q'): An irrational number is a number that cannot be expressed in the rational form p/q . The decimal representation of an irrational number is **non-terminating and non-recurring**.

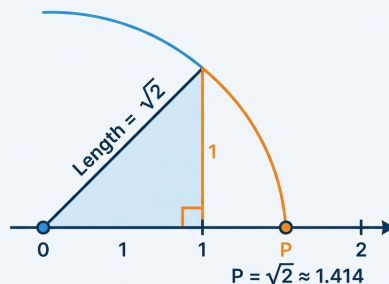
Examples:

- Square roots of non-square integers: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$.
- The mathematical constants: π (Pi $\approx 3.14159...$) and e (Euler's number $\approx 2.71828...$).
- Pattern decimals: $0.1011011101110...$ (since the number of 1s keeps increasing, it never repeats a fixed block).

Representing $\sqrt{2}$ on the Number Line:

We can locate irrational numbers like $\sqrt{2}$ and $\sqrt{3}$ on a number line geometrically using the Pythagorean theorem ($c^2 = a^2 + b^2$):

1. Take a horizontal line and mark origin O as 0, point A as 1 unit. So, $OA = 1$.
2. Construct a perpendicular AB of length 1 unit at A.
3. Join OB. By Pythagoras theorem in right triangle OAB: $OB = \sqrt{OA^2 + AB^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$.
4. Using a compass with center O and radius OB, draw an arc intersecting the number line at P. The point P represents the number $\sqrt{2} (\approx 1.414)$.



The Real Number System (R): The collection of all rational numbers and irrational numbers together forms the Real Number system. Thus, every real number is represented by a unique point on the number line, and conversely, every point on the number line represents a unique real number.

3. SURDS AND OPERATIONS

A **Surd** is an irrational root of a positive rational number. In other words, a number of the form $^n\sqrt{a}$ is a surd if a is a positive rational number and $^n\sqrt{a}$ is irrational.

In the surd expression $^n\sqrt{a}$:

- $\sqrt{}$ is called the **radical sign**.
- a is called the **radicand** (must be a positive rational number).
- n is called the **index** or **order of the surd** (must be an integer > 1).

Note: $\sqrt{9}$ is NOT a surd because $\sqrt{9} = 3$ (which is rational). But $\sqrt{8}$ is a surd because its value is irrational.

Classification of Surds

Pure vs. Mixed Surds

Pure Surd: A surd whose coefficient in rational form is 1. Examples: $\sqrt{3}$, $^3\sqrt{10}$.

Mixed Surd: A surd whose coefficient is a rational number other than 1. Examples: $2\sqrt{5}$, $5^3\sqrt{2}$. (Note: $2\sqrt{5} = \sqrt{4 \times 5} = \sqrt{20}$, which is its pure form).

Like vs. Unlike Surds

Like Surds: Surds that have the same surdy part (same index and same radicand) when simplified. Examples: $3\sqrt{2}$ and $5\sqrt{2}$.

Unlike Surds: Surds that have different radicands or different indices. Examples: $3\sqrt{2}$ and $3\sqrt{5}$.

Rationalising the Denominator & Conjugate Surds

If the denominator of a fraction contains a surd, we multiply both numerator and denominator by an appropriate term (called the **rationalising factor**) to convert the denominator into a rational number.

- For a simple surd denominator like $\sqrt{a}$, the rationalising factor is $\sqrt{a}$ (since $\sqrt{a} \times \sqrt{a} = a$).
- For a binomial surd like $a + \sqrt{b}$, the rationalising factor is its **conjugate surd**, which is $a - \sqrt{b}$.
Example: $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$, which is a rational number.

4. LAWS OF RADICALS & SCIENTIFIC NOTATION

Let a and b be positive real numbers and m, n be positive integers. The laws of radicals are:

- **Law 1 (Power of a Radical):** $(^n\sqrt{a})^n = a \Rightarrow (a^{1/n})^n = a$
- **Law 2 (Product Rule):** $^n\sqrt{a} \times ^n\sqrt{b} = ^n\sqrt{ab}$
- **Law 3 (Quotient Rule):** $^n\sqrt{a} / ^n\sqrt{b} = ^n\sqrt{a/b}$ (for $b \neq 0$)
- **Law 4 (Radical of a Radical):** $^m\sqrt{(^n\sqrt{a})} = ^{mn}\sqrt{a} = ^n\sqrt{(^m\sqrt{a})}$

Scientific Notation

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Writing very large or very small numbers is difficult and prone to errors. **Scientific Notation** is a standard way of representing any number as a decimal value between 1.0 and 10.0 multiplied by a power of 10.

$$N = a \times 10^n \quad (\text{where } 1 \leq a < 10, \text{ and } n \text{ is an integer})$$

Converting Standard to Scientific

- Large Numbers: Move decimal to the left. The exponent n is positive.

Example: $354,000 = 3.54 \times 10^5$ (moved 5 places left).

- Small Numbers: Move decimal to the right.

The exponent n is negative.

Example: $0.000028 = 2.8 \times 10^{-5}$ (moved 5 places right).

Operations in Scientific Notation

- Multiplication: Multiply coefficients and add exponents.

Example: $(2 \times 10^4) \times (3 \times 10^5) = 6 \times 10^9$.

- Division: Divide coefficients and subtract exponents.

Example: $(8 \times 10^7) / (2 \times 10^3) = 4 \times 10^4$.

5. SOLVED NUMERICAL PROBLEMS

Problem 1: Express the recurring decimal $0.2\overline{35}$ in the rational form p/q .

Solution: Let $x = 0.2353535\ldots$ --- (1)

Multiply by 10 to bring the non-repeating part to the left of the decimal:

$$10x = 2.353535\ldots \quad \text{--- (2)}$$

Multiply (2) by 100 (since the repeating block 35 has length 2):

$$1000x = 235.353535\ldots \quad \text{--- (3)}$$

Subtract (2) from (3):

$$1000x - 10x = (235.3535\ldots) - (2.3535\ldots) \Rightarrow 990x = 233$$

$$x = \frac{233}{990}$$

Problem 2: Rationalise the denominator of $\frac{5}{\sqrt{3} - 2\sqrt{2}}$.

Solution: The conjugate of the denominator $\sqrt{3} - 2\sqrt{2}$ is $\sqrt{3} + 2\sqrt{2}$.

Multiply numerator and denominator by the conjugate:

$$\frac{5}{\sqrt{3} - 2\sqrt{2}} \times \frac{\sqrt{3} + 2\sqrt{2}}{\sqrt{3} + 2\sqrt{2}} = \frac{5(\sqrt{3} + 2\sqrt{2})}{(\sqrt{3})^2 - (2\sqrt{2})^2}$$

Calculate denominator: $(\sqrt{3})^2 = 9$ and $(2\sqrt{2})^2 = 4 \times 2 = 8$.

Denominator $= 9 - 8 = 1$.

Result $= \frac{5(\sqrt{3} + 2\sqrt{2})}{1} = 5(\sqrt{3} + 2\sqrt{2})$ (or $\frac{5(\sqrt{3} + 2\sqrt{2})}{1}$).

Problem 3: Simplify: $4\sqrt{75} + 3\sqrt{48} - 2\sqrt{243}$.

Solution: Express each surd in its simplest mixed form:

$$4\sqrt{75} = 4\sqrt{25 \times 3} = 4 \times 5\sqrt{3} = 20\sqrt{3}$$

$$3\sqrt{48} = 3\sqrt{16 \times 3} = 3 \times 4\sqrt{3} = 12\sqrt{3}$$

$$2\sqrt{243} = 2\sqrt{81 \times 3} = 2 \times 9\sqrt{3} = 18\sqrt{3}$$

Now combine the like terms:

$$20\sqrt{3} + 12\sqrt{3} - 18\sqrt{3} = (20 + 12 - 18)\sqrt{3} = 14\sqrt{3}$$

Problem 4: Find the value of $(3.4 \times 10^4) \times (2.5 \times 10^{-6})$ and express the result in scientific notation.

Solution: Multiply the coefficients and add exponents:

$$(3.4 \times 2.5) \times 10^{4 + (-6)}$$

$$3.4 \times 2.5 = 8.5$$

$$\text{Exponent} = 4 - 6 = -2$$

$$\text{Result} = \mathbf{8.5 \times 10^{-2}}$$

Since 8.5 is between 1 and 10, the answer is already in standard scientific notation.

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